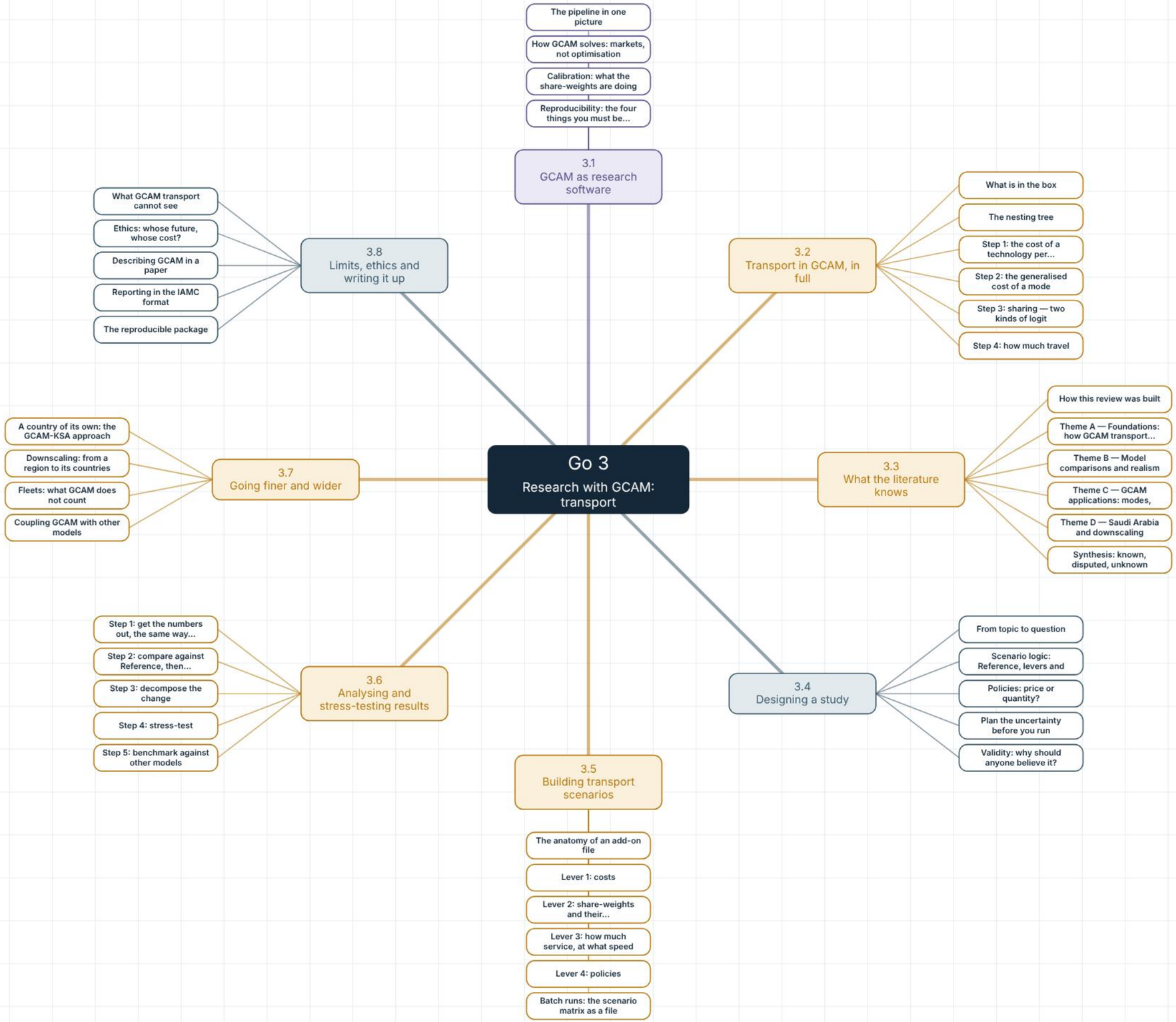
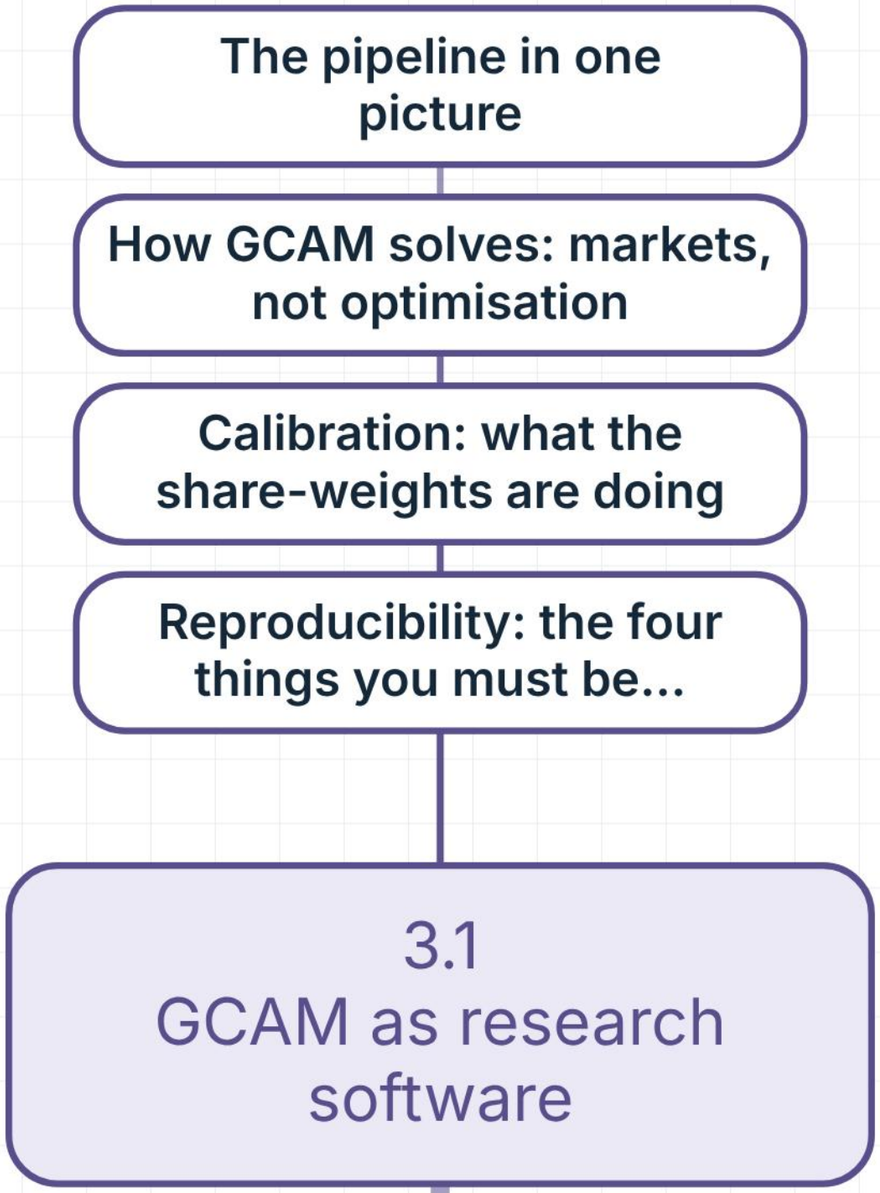


Go 3: Research with GCAM: transport

Design, build and defend a research study with GCAM, using the transport sector as the running case.

8 modules. Press Next to zoom into each one.



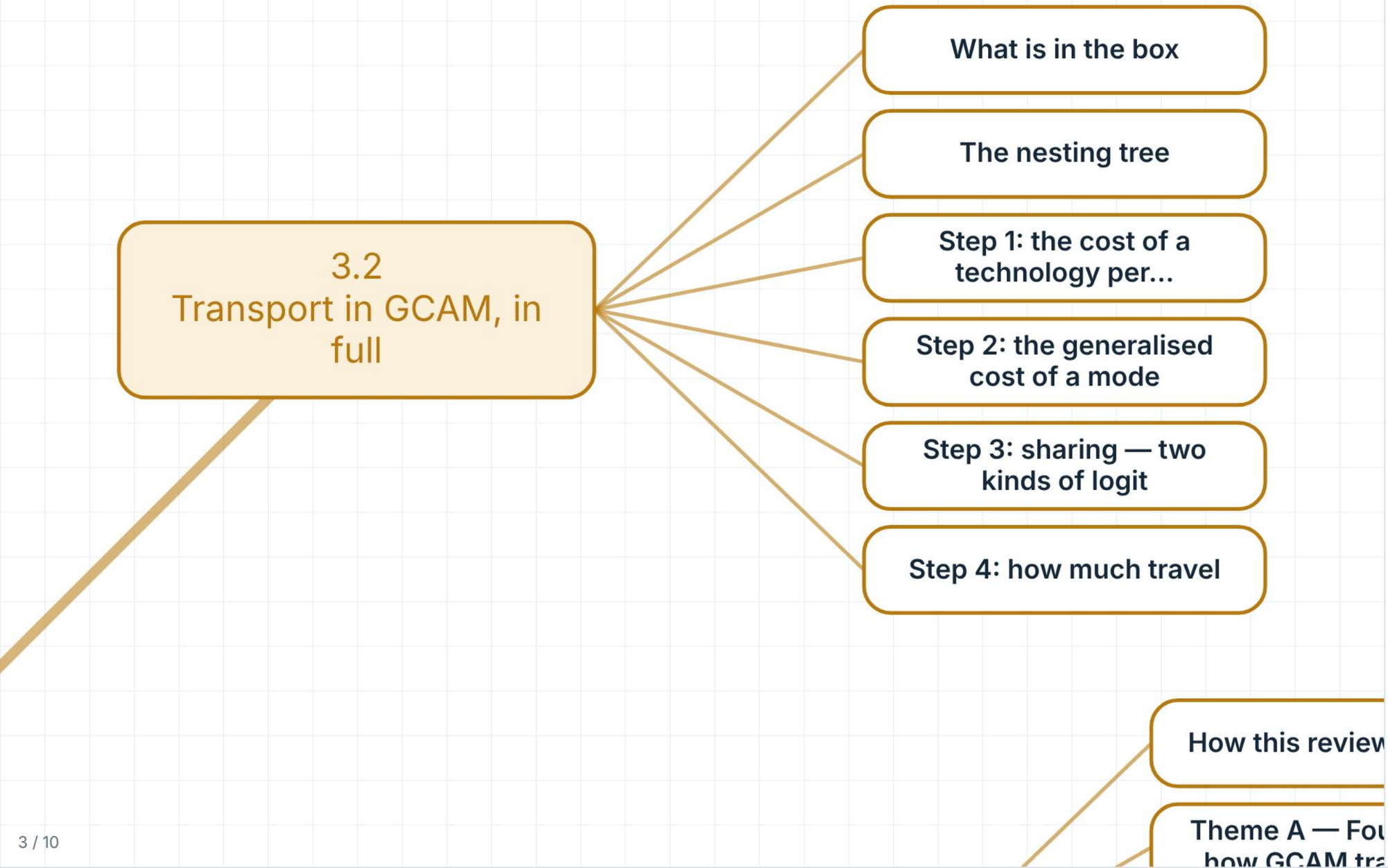
3.1 GCAM as research software

Treat GCAM as research software: know the pipeline from source data to results, how the solver works, what calibration fixes, and what you must keep to rebuild any run.

Remember

- A GCAM result is the end of a chain: source data, gcamdata, input XML, configuration, solver, database, queries
- GCAM is recursive-dynamic: it clears markets one 5-year period at a time, with no foresight
- Share-weights reproduce the base year; after that they are your assumptions and must be reported
- Record version, inputs, configuration and queries; the docs describe v8.2 while the release is 9.1

Open module 3.1



3.2 Transport in GCAM, in full

Passenger and freight transport in GCAM 9.1 in full: the nesting tree, the cost per passenger-km, the value of time, two kinds of logit, and how much travel is demanded.

Remember

- Passenger transport is a tree: service → modes → road → light vehicles → 4W → technologies
- Technology cost per pkm = (non-fuel cost + fuel price × intensity) ÷ load factor
- Mode cost adds wage × time-value multiplier ÷ speed, with wage = GDP per person ÷ 2,000 h
- Modes compete by relative-cost logits; technologies inside a mode by absolute-cost logits

Open module 3.2

Step 3: sharing — two kinds of logit

Step 4: how much travel

3.3
What the literature knows

How this review was built

Theme A — Foundations: how GCAM transport...

Theme B — Model comparisons and realism

Theme C — GCAM applications: modes,

Theme D — Saudi Arabia and downscaling

Synthesis: known, disputed, unknown

From topic to question

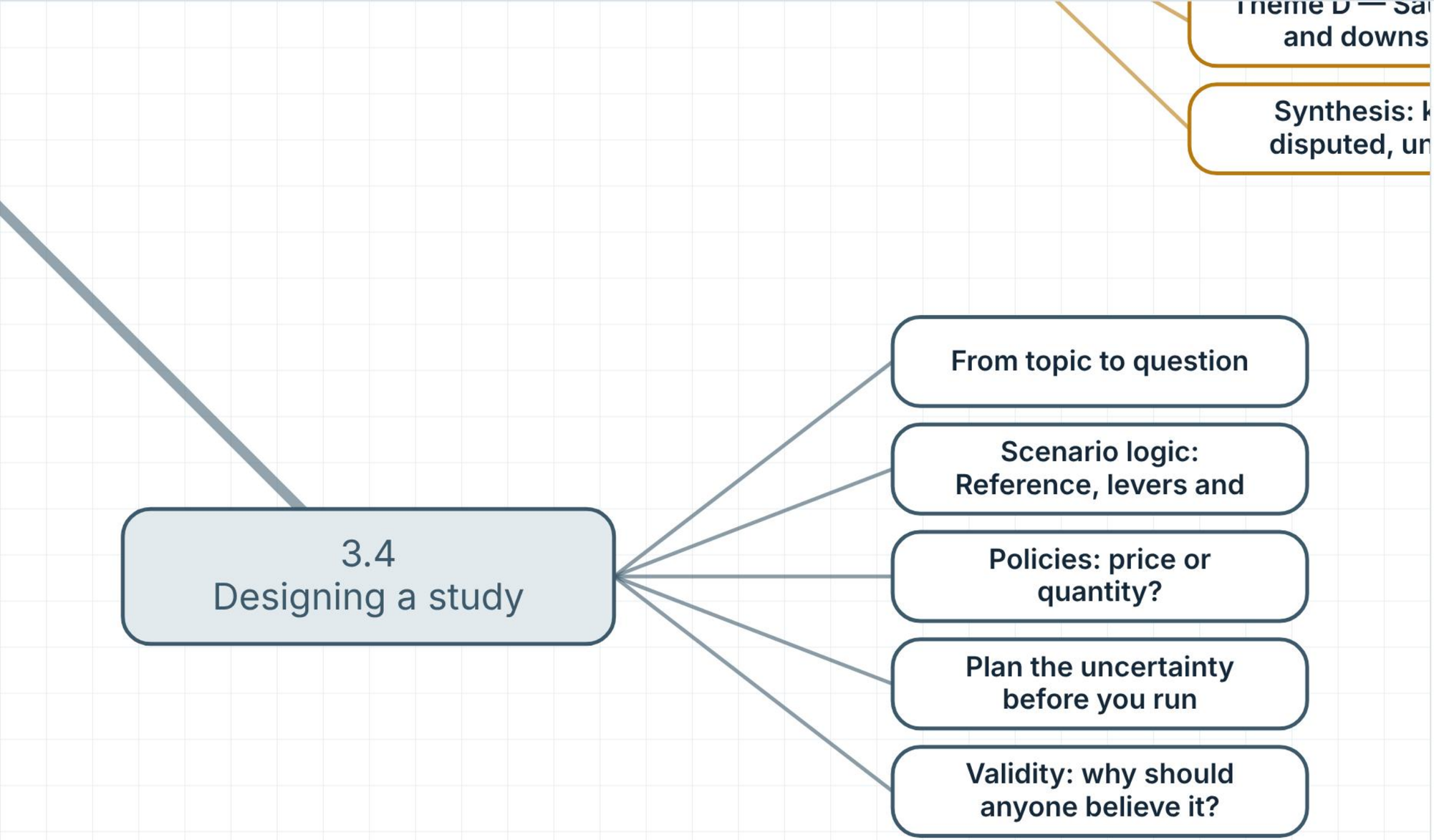
3.3 What the literature knows

A structured review of what GCAM and other integrated assessment models say about decarbonising transport: foundations, model comparisons, applications by mode and fuel, and Saudi Arabia.

Remember

- Read the four strands before choosing a question: foundations, comparisons, applications, Saudi Arabia and downscaling
- Models disagree mainly through assumptions on behaviour, costs and demand
- Electrification of light vehicles is the most robust finding; aviation, shipping and freight fuels are the least settled
- Check every DOI and detail marked unverified before citing it

Open module 3.3



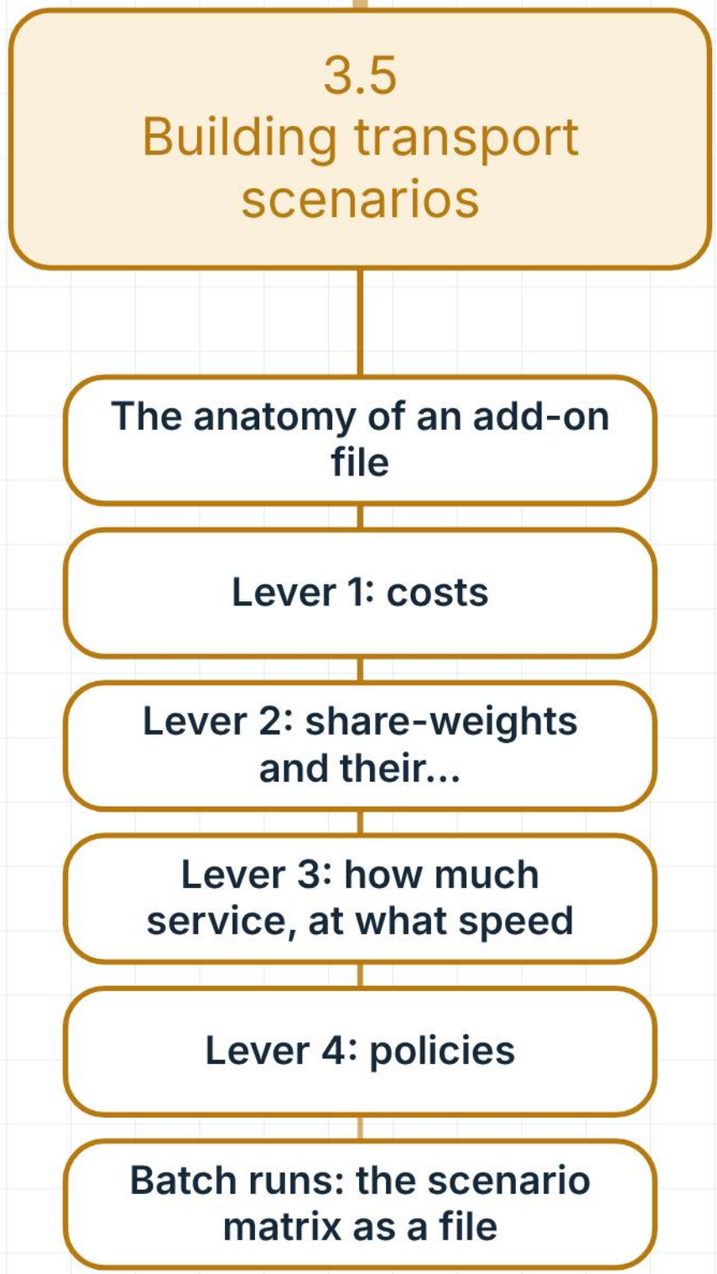
3.4 Designing a study

Turn a topic into a research question GCAM can answer, design a Reference and levers, choose price or quantity policies, and plan the uncertainty analysis before you run anything.

Remember

- A question names a decision, a metric and a comparison
- Change one lever at a time against a fully described Reference
- A price fixes cost and finds emissions; a cap fixes emissions and finds the price
- Plan inputs, ranges and method for uncertainty before the first run

Open module 3.4



3.5 Building transport scenarios

Build transport scenarios as GCAM add-on XML: costs, share-weights with interpolation rules, load factors and speeds, carbon taxes, net-zero constraints and a zero-emission-vehicle credit market, run as a batch.

Remember

- An add-on repeats GCAM's nesting down to the one value it changes, and loads after the base files
- Match units to your release: billion pkm and 1975$ in transport since v8.5
- Share-weights with interpolation rules control when new technologies can enter
- A carbon tax fixes the price; a constraint fixes emissions; a ZEV standard can be a credit market

Open module 3.5

CAM with other
models

Step 1: get the numbers out, the same way...

Step 2: compare against Reference, then...

Step 3: decompose the change

Step 4: stress-test

Step 5: benchmark against other models

3.6
Analysing and stress-testing results

3.6 Analysing and stress-testing results

Get results out the same way every time, compare against Reference, decompose changes in transport CO2, stress-test with sensitivity runs and scenario discovery, and benchmark against other models.

Remember

- Same queries, tidy tables, scripted figures: every number rebuildable
- Report policy minus Reference, never a policy run alone
- LMDI splits a CO2 change into activity, structure, intensity and carbon effects that add up exactly
- Tornado charts rank single assumptions; scenario discovery finds combinations that matter

Open module 3.6

The reproducible package

A country of its own: the GCAM-KSA approach

Downscaling: from a region to its countries

Fleets: what GCAM does not count

Coupling GCAM with other models

3.7
Going finer and wider

Step 1: get the numbers out, the same way...

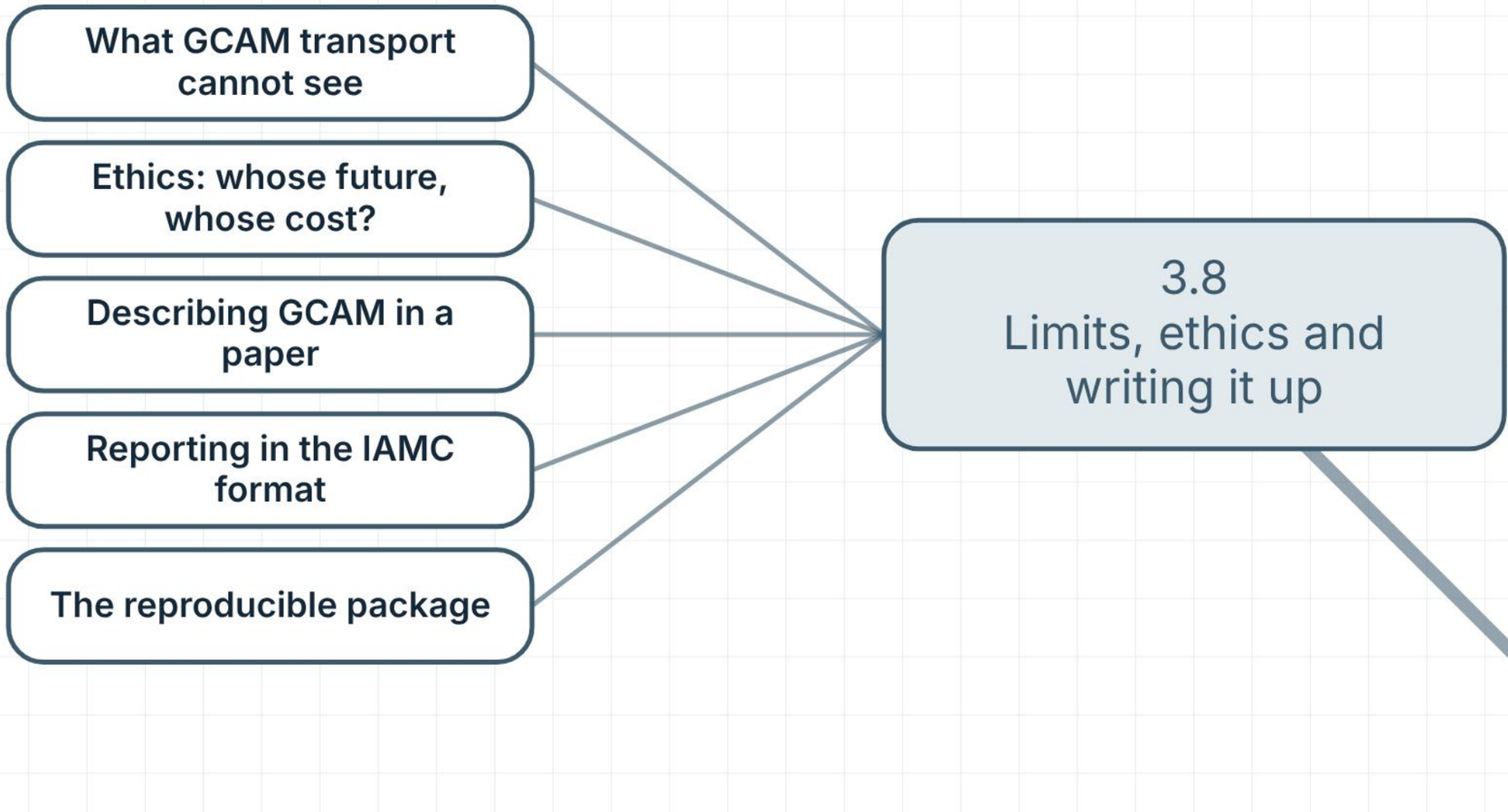
3.7 Going finer and wider

Go finer by giving a country its own region (the GCAM-KSA approach) or downscaling regional results to countries, and go wider by coupling GCAM with vehicle-stock and national energy models.

Remember

- Two ways to go finer: a country region (GCAM-KSA) or downscaling regional results
- Downscaled countries must still add up to the regional total; state the convergence assumption
- GCAM models sales and service, not fleets: use a stock model for vehicles on the road
- Coupling is about which variables pass each way and iterating to agreement

Open module 3.7



3.8 Limits, ethics and writing it up

Know what GCAM transport cannot see, weigh whose future and whose cost your scenarios assume, describe GCAM properly in a paper, report in IAMC format, and publish a reproducible package.

Remember

- Name GCAM transport's blind spots and how they could bias your answer
- Make the ethical choices in your scenarios visible
- State the exact version and every change, following the GCAM community guidelines
- Report in IAMC variables and publish a package that rebuilds every figure

Open module 3.8

of its own: the
KSA approach